Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data columns tracked

37

Missing Values

In review ratings

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Data Preparation Process

01

Data Loading

Imported using pandas

02

Initial Exploration

Structure and statistics check

03

Missing Data Handling

Imputed median ratings by category

04

Feature Engineering

Created age groups and purchase frequency

05

Database Integration

Loaded to PostgreSQL for SQL analysis

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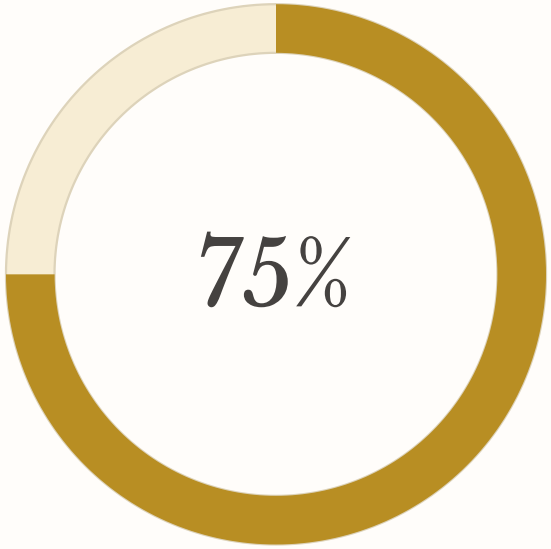
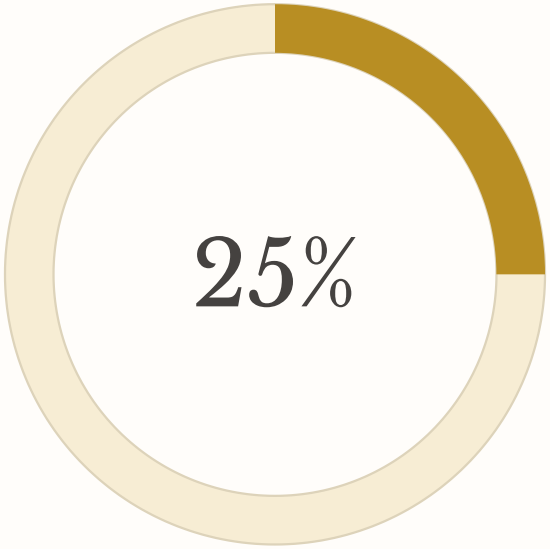
Revenue by Gender



Male Customers

$157,890 revenue



Female Customers

$75,191 revenue

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Top Products by Rating

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

3.80 average rating

5

Skirt

3.78 average rating

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Shipping & Subscription Impact

Shipping Type

Standard: $58.46 avg

Express: $60.48 avg

Subscription Status

Subscribers: 1,053 customers

Non-subscribers: 2,847 customers

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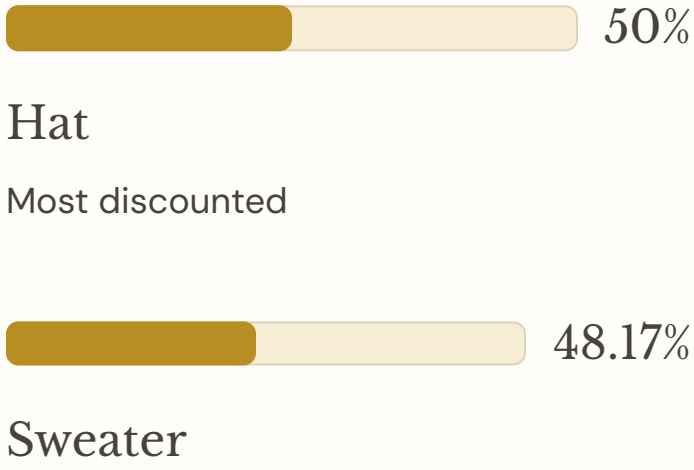
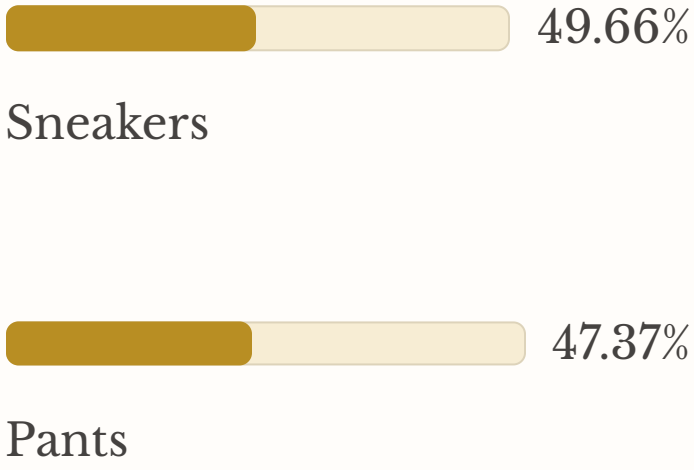
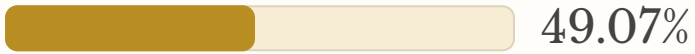
Discount-Dependent Products







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Customer Segmentation

Loyal

3,116 customers

Returning

701 customers

New

83 customers

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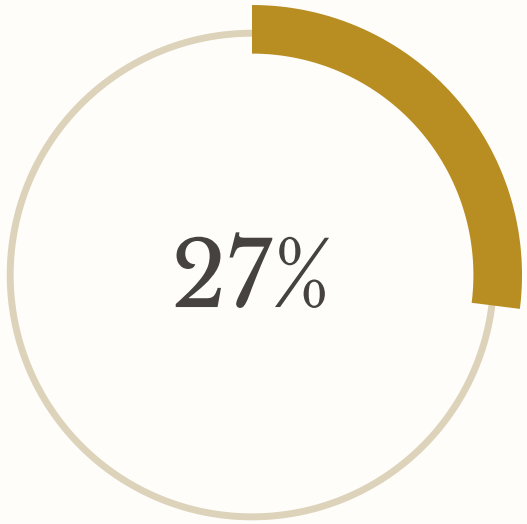
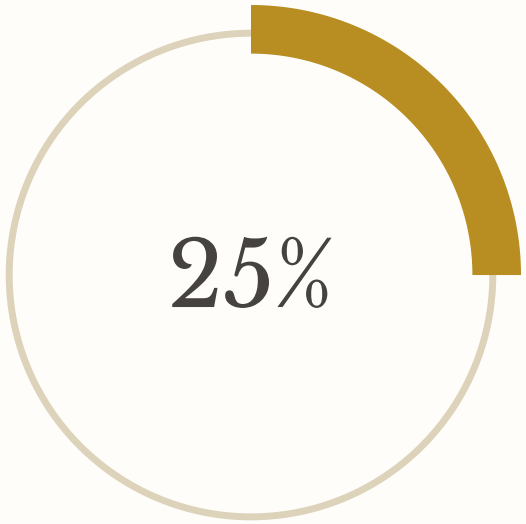
Revenue by Age Group



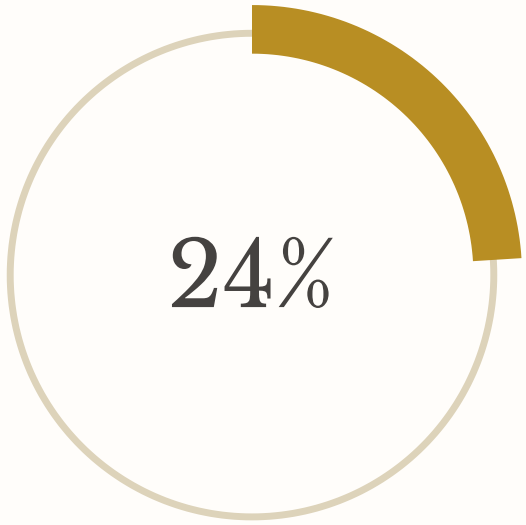
Young Adult

$62,143 revenue



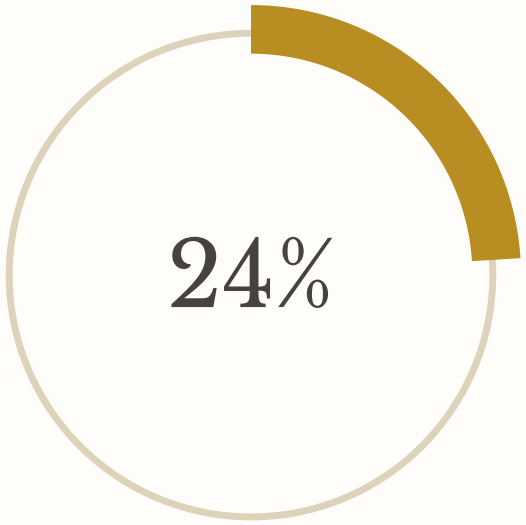
Middle-aged

$59,197 revenue



Adult

$55,978 revenue



Senior

$55,763 revenue

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits

Loyalty Programs

Reward repeat buyers

Review Discount Policy

Balance sales with margins

Targeted Marketing

Focus on high-revenue groups

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